

Fixed-Edit FIBER–GRAND: Flagship-Paper Claim

Contract and Readiness Gate

Exact aggregate inverse decoding, membership-oracle query optimality, moment/tail laws, and a same-toolchain compiled benchmark

Scientific development and de-risking dossier, version 4.0

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Scientific status after gates v1–v3

The broad FIBER–GRAND proposal must remain narrowed. Posterior candidate rank is ordinary conditional guesswork; best-path versus best-string inference, hidden-path summation, priority-first search, and generic candidate-likelihood guessing are established. The surviving programme concerns a fixed number of synchronization edits and a weak code interface.

The v3 WSL reproduction passed 18 implementation tests and returned a high-potential but not-yet-established classification. That result supports continued research, but it is not yet a quality-flagship-paper verdict: the largest benchmark used only three blocks per configuration and Python timing remained a confound.

Version 4 adds a decoder-specific membership-oracle lower bound, a matching candidate-query exponent for FIBER, and a same-toolchain compiled C++20 comparison. Its positive classification authorizes preparation of a full IEEE Transactions on Information Theory manuscript and independent external proof/novelty review. It still does not establish field-defining impact.

Contents

1	Authoritative verdict on the v3 evidence	3
2	Permanent novelty boundary	3
3	Channel model and exact membership-first decoder	4
4	New candidate-volume and membership-oracle theory	4
4.1	Candidate shells	4
4.2	The L0 oracle model	6
4.3	Query-competitiveness theorem	6
5	Fixed-edit history moments, tails, and phase boundaries	7
6	Same-toolchain compiled flagship benchmark	7
7	Specialization and real-system boundaries	8
8	Flagship manuscript contract	8
9	Frozen decision rules	9

10 External-review checklist	9
11 Conclusion	9

1 Authoritative verdict on the v3 evidence

The v3 computation established a coherent theory-and-algorithm core:

- (i) exact complete-tie agreement on exhaustive micro-worlds;
- (ii) a codebook-free membership-only decoder using syndromes or checksums;
- (iii) a strict unseen-candidate certificate;
- (iv) a deterministic fixed-edit shell upper bound;
- (v) history-revelation moment and tail exponents inherited from BSC guesswork;
- (vi) a typical-versus-mean phase diagram;
- (vii) finite-length wins over exact prefix and syndrome-trellis searches in low-substitution regimes.

The same evidence also revealed hard boundaries. At rate $R = 0.75$, the theoretical mean-work boundary is near $p = 0.009$, much lower than the fixed-quantile boundary near $p = 0.0417$. The v3 $p = 0.04$ rows displayed severe search tails, and the generic decoder was much slower than a true linear-time VT specialization. The programme is therefore not a general insertion/deletion decoder; it is a fixed-edit, low-substitution, high-rate candidate.

The v3 computational decision was intentionally permissive. Three observations prevent immediate manuscript freeze:

- (a) the largest length was $n = 40$ with only three trials per row;
- (b) percentile labels from three observations are not statistically meaningful tail estimates;
- (c) Python implementations can favor one search organization because of array allocation, interpreter, and dynamic-dispatch costs.

Version 4 addresses these deficiencies before manuscript investment.

2 Permanent novelty boundary

The following claims are prohibited.

1. *Fiber guesswork is a new random variable.* Ranking X by $W(Y | X)$ under a uniform prior is ordinary conditional guesswork [4, 5].
2. *Best path versus best aggregate candidate is new.* The distinction is established in probabilistic automata and weighted inference [10, 9].
3. *Priority queues, A^* , or path summation are new.* Priority-first ML decoding and synchronization sequential decoding are classical lines [6, 7, 8].
4. *General channel candidate guessing is new.* Guessing-based decoding for general DMCs and unified GRAND/GCD formulations already exist [2, 3].
5. *Efficient exact decoding holds for unrestricted weighted transducers.* Best-string inference can be hard, and the current algorithm is XP rather than FPT in edit count [10, 11].

The narrow residual claim is:

For a fixed number of uniformly located deletions followed by survivor BSC substitutions, a codebook-free membership-only aggregate inverse decoder admits a strict exact-ML certificate. Its distinct-candidate query count is exponent-optimal within an explicit L0 membership-oracle model, its history work obeys fixed-edit moment and tail laws, and a hybrid with code-side exact search has distinct typical and mean operating boundaries.

No located primary source contains this complete theorem-and-algorithm package. Several ingredients are known separately, including fixed-deletion multiplicity-aware likelihoods [13], edit-frame enumeration in an earlier patent [14], and specialized deletion/substitution code constructions [15, 16]. The correct novelty label is *distinct but narrow*, subject to independent external adjudication.

3 Channel model and exact membership-first decoder

Let $x \in \{0, 1\}^n$. Exactly $t < n$ positions are selected uniformly without replacement and deleted. The $m = n - t$ survivors pass through a BSC with crossover probability $0 \leq p < 1/2$, producing $y \in \{0, 1\}^m$. Let

$$\mathcal{J}_t = \{J \subseteq [n] : |J| = t\}, \quad S = |\mathcal{J}_t| = \binom{n}{t}.$$

For $J \in \mathcal{J}_t$, let $d_J(x, y)$ be the Hamming distance between y and the subsequence of x obtained by deleting J . The exact likelihood is

$$\Lambda_y(x) = \frac{1}{S} \sum_{J \in \mathcal{J}_t} p^{d_J(x, y)} (1 - p)^{m - d_J(x, y)}. \quad (1)$$

An inverse history is $h = (J, b, e)$, where $b \in \{0, 1\}^t$ fills the deleted symbols and $e \in \{0, 1\}^m$ is the survivor substitution pattern. It creates

$$x_h = \text{Ins}_J(y \oplus e, b)$$

with component weight

$$\omega(h) = S^{-1} p^{\text{wt}(e)} (1 - p)^{m - \text{wt}(e)}.$$

For each J and complete x , there is exactly one compatible (b, e) .

The decoder maintains one probability-ordered stream for every J . At first discovery of a distinct x , it queries $\text{IsCodeword}(x)$. It evaluates the complete sum (1) only for codewords. If η_J is the next unprocessed component weight in stream J , then every completely unseen candidate satisfies

$$\Lambda_y(x) \leq U_{\text{new}} := \sum_{J \in \mathcal{J}_t} \eta_J. \quad (2)$$

If the largest exactly evaluated codeword score s^* satisfies $s^* > U_{\text{new}}$, the complete discovered tie set at s^* is the global ML tie set.

4 New candidate-volume and membership-oracle theory

This section gives the main new contribution of version 4. Unlike the history-revelation theorem, it directly compares the realized decoder with the best possible exact decoder under the same weak code interface.

4.1 Candidate shells

Define the best-alignment mismatch

$$\delta_y(x) := \min_{J \in \mathcal{J}_t} d_J(x, y)$$

Algorithm 1 Membership-first certified FIBER search

Require: y, t, p , membership oracle for $\mathcal{C}$

- 1: initialize one probability-ordered inverse stream for every $J \in \mathcal{J}_t$
- 2: $D \leftarrow \emptyset, s^* \leftarrow -\infty, T^* \leftarrow \emptyset$
- 3: **while** some stream remains **do**
- 4: pop the globally largest next history and advance its stream
- 5: construct x_h
- 6: **if** $x_h \notin D$ **then**
- 7: $D \leftarrow D \cup \{x_h\}$ and query $\text{IsCodeword}(x_h)$
- 8: **if** $x_h \in \mathcal{C}$ **then**
- 9: compute exact $\Lambda_y(x_h)$ and update (s^*, T^*)
- 10: **end if**
- 11: **end if**
- 12: $U_{\text{new}} \leftarrow \sum_J \eta_J$
- 13: **if** $T^* \neq \emptyset$ and $s^* > U_{\text{new}}$ **then**
- 14: **return** T^* with strict ML certificate
- 15: **end if**
- 16: **end while**

and the inverse candidate-shell volume

$$V_y(w) := |\{x \in \{0, 1\}^n : \delta_y(x) \leq w\}|.$$

Let

$$B_m(w) = \sum_{i=0}^{\min\{w, m\}} \binom{m}{i},$$

with $B_m(w) = 0$ for $w < 0$.

Proposition 1 (Candidate-volume and ambiguity bounds). *For every observation y and integer w ,*

$$2^t B_m(w) \leq V_y(w) \leq 2^t \binom{n}{t} B_m(w). \quad (3)$$

Moreover, every complete candidate $x \in \{0, 1\}^n$ has exactly $\binom{n}{t}$ compatible inverse histories.

Proof. Fix one deletion set J . The map $(e, b) \mapsto \text{Ins}_J(y \oplus e, b)$ is injective, so the histories with $\text{wt}(e) \leq w$ create exactly $2^t B_m(w)$ distinct candidates. This proves the lower bound. Taking the union over the $S = \binom{n}{t}$ deletion sets proves the upper bound. For fixed (x, J) , the deleted bits b are the symbols of x in J , and $e = y \oplus \text{Del}_J(x)$ is unique. Hence there is one compatible history per J . $\square$

Define the largest individual component supporting x :

$$M_y(x) = \frac{1}{S} p^{\delta_y(x)} (1-p)^{m-\delta_y(x)}.$$

Then

$$M_y(x) \leq \Lambda_y(x) \leq S M_y(x). \quad (4)$$

4.2 The L0 oracle model

Definition 1 (L0 membership-oracle decoder). An L0 decoder knows the channel and observation and may perform arbitrary channel-side computation, but it learns the code only through adaptive queries of the form $\text{IsCodeword}(x)$. It must return an exact ML decision for every membership-testable codebook $\mathcal{C} \subseteq \{0, 1\}^n$.

Lemma 1 (Mandatory higher-likelihood queries). *Before certifying a codeword c^* , every exact L0 decoder must query every candidate x satisfying*

$$\Lambda_y(x) > \Lambda_y(c^*).$$

Proof. If such an x remains unqueried, the transcript is consistent with another membership oracle that agrees on every previous query but also declares x a codeword. Under that oracle, c^* is not ML. Thus the original transcript cannot certify an exact answer for every codebook. $\square$

4.3 Query-competitiveness theorem

Let c^* be an ML codeword and write $d^* = \delta_y(c^*)$. For $p > 0$, set

$$a = \frac{1-p}{p}, \quad L_{n,t} = \left\lfloor \log_a \binom{n}{t} \right\rfloor.$$

Let Q_{L0}^* be the minimum number of distinct membership queries made by any exact L0 decoder on this instance, and let Q_{FIBER} be the number made by FIBER.

Theorem 1 (Oracle-relative candidate-query sandwich). *For $0 < p < 1/2$,*

$$V_y(d^* - L_{n,t} - 1) \leq Q_{\text{L0}}^* \leq Q_{\text{FIBER}} \leq V_y(d^* + L_{n,t}). \quad (5)$$

At $p = 0$, the same argument gives a polynomial fixed-edit bound after shell zero.

Proof. If $\delta_y(x) \leq d^* - L_{n,t} - 1$, then by (4),

$$\Lambda_y(x) \geq M_y(x) \geq \frac{1}{S} p^{d^* - L_{n,t} - 1} (1-p)^{m - d^* + L_{n,t} + 1} > p^{d^*} (1-p)^{m - d^*} \geq \Lambda_y(c^*),$$

because $a^{L_{n,t}+1} > S$. Lemma 1 proves the first inequality. The middle inequality is the definition of an optimal L0 query count. FIBER discovers c^* by shell d^* . After all histories through shell $d^* + L_{n,t}$ have been emitted, the sum-of-frontiers bound is strictly below the probability of one best supporting history of c^* , and therefore below $\Lambda_y(c^*)$. The certificate has fired, and every queried candidate lies in $V_y(d^* + L_{n,t})$. $\square$

Corollary 1 (Subexponential and interior-shell polynomial competitiveness). *For fixed t , if $d^*/m \rightarrow q \in (0, 1/2)$, then*

$$\frac{1}{n} \log_2 Q_{\text{L0}}^* = \frac{1}{n} \log_2 Q_{\text{FIBER}} + o(1) = h_2(q) + o(1),$$

and

$$\frac{Q_{\text{FIBER}}}{Q_{\text{L0}}^*} \leq n^{O(1)}.$$

Thus FIBER is candidate-query exponent-optimal in the L0 oracle model on every such instance sequence.

Proof. For fixed t , $L_{n,t} = O(\log n)$ and $S = n^{O(1)}$. Standard Hamming-ball estimates applied to (3) give

$$\log_2 B_m(d^* \pm O(\log n)) = m h_2(q) + O(\log n).$$

Apply Theorem 1. $\square$

Remark 1 (Scope). The lower bound is an oracle result, not a processor-cycle converse. A decoder exploiting a parity-check matrix, trellis, or algebraic completion oracle is outside L0 and can be faster. This is why the experimental gate compares FIBER with code-aware exact searches and retains the VT specialization boundary.

5 Fixed-edit history moments, tails, and phase boundaries

Let R_{hist} be the probability-order rank of the actual inverse history. With

$$K_n = 2^t \binom{n}{t}, \quad B_m(k) = \sum_{i=0}^{\min\{k,m\}} \binom{m}{i},$$

and actual survivor error weight $E = e$,

$$1 + K_n B_m(e - 1) \leq R_{\text{hist}} \leq K_n B_m(e).$$

For fixed t , $0 < p < 1/2$, and $\rho > 0$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{E}[R_{\text{hist}}^\rho] = \rho H_{1/(1+\rho)}(p), \quad (6)$$

and the actual certified FIBER work obeys the corresponding upper moment bound. Equality in (6) concerns history revelation, not every code-dependent stopping time.

For $\gamma \in [h_2(p), 1)$, let $q_\gamma \in [p, 1/2)$ solve $h_2(q_\gamma) = \gamma$. Then

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \Pr \{n^{-1} \log_2 R_{\text{hist}} \geq \gamma\} = D_2(q_\gamma \| p).$$

Every fixed quantile has exponent $h_2(p)$, while the mean exponent is $H_{1/2}(p)$.

If an independent code-side exact decoder uses at most $\text{poly}(n)2^{n(1-R)}$ work, a parallel hybrid has upper-envelope boundaries

$$\min\{h_2(p), 1 - R\} \quad \text{for fixed quantiles,}$$

and

$$\min\{H_{1/2}(p), 1 - R\} \quad \text{for mean work.}$$

The resulting thresholds are:

R	$1 - R$	$p_{\text{typ}}: h_2(p) = 1 - R$	$p_{\text{mean}}: H_{1/2}(p) = 1 - R$
0.625	0.375	0.07245	0.02254
0.750	0.250	0.04169	0.00903
0.875	0.125	0.01713	0.00205

These are exponent boundaries, not finite-length guarantees.

6 Same-toolchain compiled flagship benchmark

The version-4 benchmark compiles both FIBER and its exact code-aware references in one C++20 translation unit using the same compiler, optimization flags, channel likelihood, code representation, and timing clock. Decoder execution order is randomized for each block. The benchmark includes:

1. a membership-only FIBER history search;
2. a systematic-code prefix A^* search with an admissible aggregate likelihood bound;
3. a syndrome-trellis aggregate A^* reference when its dynamic-programming state budget is feasible;
4. exhaustive small-world self-tests against ML.

At the largest lengths, compiled prefix A* is the authoritative timed reference. The syndrome-trellis decoder is timed only where its dynamic-programming state budget is feasible and is independently exercised against exhaustive ML in the compiled self-test. Random-linear and reproducible CRC-polynomial codes use syndrome membership without materialized codebooks.

The standard profile tests $n \in \{32, 48, 64\}$, rates 0.75 and 0.875, and probabilities concentrated around the theoretically favorable region. It reports exact agreement, completion, median ratio with bootstrap confidence interval, 90/95/99th observed ratios, raw histories, membership queries, prefix/trellis states, and fitted scaling slopes. Boundary configurations are retained even when FIBER loses.

The gate is deliberately stricter than v3:

- all exact decoders must certify and agree;
- at least four largest-length favorable configurations must show a median $\geq 1.25\times$ FIBER speedup;
- at least three must have a bootstrap upper confidence endpoint below parity;
- at least three must have observed 95th-percentile ratio at most 1.25;
- favorable scaling may not systematically degrade;
- every mean-favorable row must remain within a 1.25 median ratio.

No finite trial count can certify a 99.9 percentile. A positive gate only establishes manuscript potential and defines the larger publication campaign.

7 Specialization and real-system boundaries

A true linear-time VT decoder is expected to dominate generic FIBER on the noiseless single-deletion VT problem. This is a positive scientific boundary: FIBER's value is code modularity under noisy ambiguous edits, not replacement of an optimal specialized decoder.

A measured post-front-end trace is optional for a theory-centered IEEE TIT manuscript. It is mandatory for statements about timing slips in deployed receivers, telecommunications value, or field-defining system impact. The package accepts measured CSV traces and audits whether they occupy the bounded-edit, low-substitution region; it does not relabel synthetic Monte Carlo data as measurement.

8 Flagship manuscript contract

A positive version-4 run authorizes a full manuscript with the following core.

1. Exact fixed-edit likelihood and membership-first aggregate inverse decoder.
2. Strict complete-tie ML certificate.
3. Candidate-volume and exact ambiguity theorem.
4. L0 membership-oracle lower bound and FIBER query-competitiveness theorem.
5. History moment and large-deviation specialization, with correct equality/upper-bound boundary.
6. Typical-versus-mean phase diagram and hybrid decoder interpretation.
7. Exact micro-world verification and same-toolchain compiled finite-length results.
8. Honest negative regimes, VT specialization boundary, and XP-not-FPT statement.

The paper must not claim universal synchronization decoding, generic fiber-guesswork novelty, FPT edit dependence, or state-of-the-art system superiority. An external reviewer must independently verify every theorem and the primary-source novelty matrix before submission.

9 Frozen decision rules

PROCEED_TO_FLAGSHIP_TIT MANUSCRIPT AND EXTERNAL REVIEW

All mathematical audits pass and the same-toolchain compiled low-substitution benchmark satisfies the preregistered robust-signal criteria. Prepare the manuscript immediately. Field-defining status remains unestablished.

NARROW_TO_STRONG_FIXED_EDIT_THEORY_AND_ALGORITHM_PAPER

The theory survives but the compiled finite-length advantage does not. Publish the exact decoder and theorem boundary without a flagship performance claim.

STOP_FIBER_FLAGSHIP_TRACK

A correctness, oracle theorem, moment/tail, or novelty defect is found. Do not rescue the claim by changing the interface after observing the result.

10 External-review checklist

The independent review must answer:

1. Is Theorem 1 valid for the stated arbitrary-code L0 oracle model?
2. Does Corollary 1 require any additional uniformity assumptions on Hamming-ball estimates?
3. Is the complete-tie certificate strict and numerically defensible?
4. Does any primary source already contain the complete candidate-volume/oracle-competitiveness result?
5. Are the moment and tail equalities correctly restricted to history revelation?
6. Are the compiled references exact and fairly charged?
7. Are all state-of-the-art and specialization boundaries represented honestly?

A blocking defect in Questions 1–5 is an immediate stop or major narrowing event.

11 Conclusion

The v3 evidence justifies continued work but not a flagship verdict by itself. The strongest surviving path is no longer a broad new decoding field. It is a precise fixed-edit theorem-and-algorithm package: exact aggregate inverse search through a weak membership interface, an auditable ML certificate, oracle-relative candidate-query optimality, and sharply different typical and mean complexity regions.

If version 4 reproduces the mathematical audits and a robust compiled advantage in the predicted low-substitution regime, a quality IEEE TIT flagship candidate is scientifically justified. If the compiled gate fails, the theory remains potentially publishable, but the flagship-performance objective must be narrowed. Real-system and field-defining claims remain conditional on measured receiver evidence and independent adoption.

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